

POKHARA UNIVERSITY

Level: Master
Semester: Fall
Programme: ME Computer
Course: Embedded Systems Design (New)

Year : 2025
Full Marks : 100
Pass Marks : 60
Time : 4 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. (a) During peak traffic on a rainy day, a vehicle equipped with ABS encounters sudden braking at 80 km/h. The road surface is slippery, and the system must adapt to prevent skidding while ensuring stopping efficiency. Propose an ABS embedded system design for this scenario, highlighting its key components and their roles. 7
- (b) Embedded Systems are everywhere, but you rarely see them. Justify this statement with suitable example. 8
2. (a) Compare and contrast DSPs and Microcontrollers in the context of embedded system use cases. Highlight scenarios where each would be more suitable. 7
- (b) Discuss about AMBA bus architecture standards: AHB and APB. Explain their respective roles in system integration and performance optimization within embedded systems. Provide a block diagram to illustrate a typical SoC bus hierarchy. 8
3. (a) Explain the concept of priority inversion in an RTOS. Illustrate how it can occur and critically evaluate the effectiveness of priority inheritance and priority ceiling protocols in mitigating it. 7
- (b) Consider a real-time embedded system controlling a robotic arm, which performs three periodic tasks. 8

Task	Execution Time (C)	Period (T)	Deadline (D)
T1 - Sensor Sampling	1 ms	4 ms	4 ms
T2 - Motor Control	1.5 ms	5 ms	5 ms
T3 - Telemetry Logging	2 ms	10 ms	10 ms

- i. Check whether these tasks are schedulable using Earliest Deadline First (EDF) Scheduling.

- ii. If yes, apply Rate Monotonic Scheduling and Earliest Deadline First Scheduling to schedule these tasks.

4. a) A rural farming cooperative deploys an IoT-based smart agriculture system with battery-powered sensor nodes, a local edge gateway, and cloud analytics. As an embedded systems engineer, propose a suitable Cloud-Edge-Device architecture for this application. Explain the roles of each layer and how data flows between them. 8
- b) Explain in detail about Bare Metal Programming along with startup code and linker script used in ARM Cortex-M3 programming. Highlight memory mapping and initialization steps. 7

OR

Explain the interrupt masking mechanism using BASEPRI, PRIMASK, and FAULTMASK with example scenarios.

5. a) Evaluate the statement: "Good embedded software is not just about meeting functional requirements, but also about maintainability." Discuss the role of modularity, abstraction, and version control in achieving long-term software maintainability. 7
- b) How design patterns improve embedded software quality? Using the Finite State Machine (FSM) pattern, explain how it simplifies the implementation of a complex control system compared to a monolithic procedural approach. 8
6. a) Describe the data transmission sequence in I2C from start to stop condition. Explain common sleep modes in low power design. 7
- b) Explain different Driver Testing and Validation Techniques. What is the role of a low-level driver in embedded systems? Explain how an ISR improves efficiency and power consumption. 8
7. Write short notes on: (Any two) 2×5
- a) IOT security basics
 - b) Unit Testing
 - c) UART and SPI Protocol
 - d) Exception types

POKHARA UNIVERSITY

Level: Master

Semester: Fall

Year : 2025

Programme: ME/MSc Computer

Full Marks : 100

Course: Artificial Intelligence (New)

Pass Marks : 60

Time : 4 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define an intelligent agent. Discuss different types of agents with example. 7
- b) What is Simulated Annealing in local search? Explain its working mechanism and how it helps in escaping local minima. 8
2. a) What is Knowledge? How can we represent it? Discuss about the Properties of a Good Knowledge representation System. 7
- b) A city uses a fuzzy logic-based traffic control system to adjust green light duration depending on traffic density. 8
 - i. How can we design Fuzzy Inference System to solve this problem?
 - ii. Explain the role of defuzzification will help in final decision-making.

OR

What are Hidden Markov Models (HMMs)? Explain their structure and how they are used for probabilistic reasoning over time with a suitable example.

3. a) Differentiate between hierarchical planning and probabilistic planning. 7
- b) Why do we require Multi-agent Planning in AI? Give an overview of Multiagent System Architecture and discuss how agent can work in Cooperative & Competitive environment. 8
4. a) What is Constraint Propagation? Explain how the constraint satisfaction problems are resolved with a suitable example. 8
- b) What do you understand by Artificial Neural network? Design a neural network and explain how it works. 7
5. a) What are the challenges of RNNs? Discuss about the architecture to solve these challenges. 7

- b) Explain the transformer model architecture. 8
6. a) Explain the steps involved in Natural language processing. 7

OR

Suppose you are hired by Prime ministers' office of Nepal to analyze the complains posted in hello sarkar portal and develop a chat bot to answer the complains posted by public.

- i. What do you think will be the issues with collected text complains.
 - ii. Discuss about the steps of NLP to solve this problem.
- b) Large language models (LLMs) have revolutionized the way we interact with AI — from chatbots like GPT-4, Gemini etc. What is the foundational innovation and architecture beneath these breakthroughs that made it all possible? Explain. 8
7. Write short notes on: (Any two) 2×5
- a) Perceptron
 - b) Transfer Learning
 - c) Complication of Real Natural Language